

Weighted Sum-Rate Maximization using Weighted MMSE for MIMO-BC Beamforming Design

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I. Introduction

II. System Model and Objective

III. Gradient Expressions & KKT Conditions

IV. Alternating Optimization

V. Simulation

I. Introduction

- For MIMO-BC, non-linear techniques outperform linear ones.
 - However, they are complex for both Tx and Rx.
- Focus on Tx Beamforming design to **Maximize Weighted Sum-Rate (WSR)**.
 - WSR maximization is non-convex and non-trivial.
- Establish a relationship between WSR maximization and **Weighted Minimum Mean Square Error (WMMSE)** minimization.
- Propose an iterative algorithm to solve WSR maximization via WMMSE minimization.

General narrow band point-to-multipoint MIMO system.

- P transmit antennas
- K users each with Q receive antennas
- System has total QK receive antennas
- MIMO Channel between Tx and user k : $\mathbf{H}_k \in \mathbb{C}^{[Q \times P]}$

$$\mathbf{y}_k(n) = \mathbf{H}_k \mathbf{x}(n) + \mathbf{v}_k(n), \quad (1)$$

Where

- $\mathbf{x}(n) \in \mathbb{C}^{[1 \times P]}$: complex transmitted vector
- $\mathbf{v}_k(n) \in \mathbb{C}^{[Q \times 1]}$: noise vector containing circularly symmetric white Gaussian noise

$\mathbf{x}(n)$: linearly filtered version of the input data vectors $\mathbf{d}_1(n), \dots, \mathbf{d}_K(n) \in \mathbb{C}^{[Q \times 1]}$

$$\mathbf{x}(n) = \sum_{k=1}^K \mathbf{B}_k \mathbf{d}_k(n). \quad (2)$$

Matrices $\mathbf{B}_1, \dots, \mathbf{B}_K \in \mathbb{C}^{[P \times Q]}$: transmit filters(Beamformers). Assumed

- each user has Q parallel data streams.
- each user receives Q independent streams such that $\mathbb{E}[\mathbf{d}_k(n)\mathbf{d}_k(n)^H] = \mathbf{I}_Q$

Transmit power constraint

$$\frac{1}{N} \sum_{n=1}^N \mathbf{x}^H(n) \mathbf{x}(n) \leq E_{tx} \quad (3)$$

If N is large enough, then

$$\frac{1}{N} \sum_{n=1}^N \mathbf{x}^H(n) \mathbf{x}(n) \rightarrow \mathbb{E}[\mathbf{x}^H(n) \mathbf{x}(n)] = \sum_k \text{Tr}(\mathbf{B}_k \mathbf{B}_k^H) \quad (3a)$$

► Derivation

To find $\mathbf{B}_1, \dots, \mathbf{B}_K$ that maximizes WSR under power constraint (3):

$$\begin{aligned} [\mathbf{B}_1^{\text{WSR}}, \dots, \mathbf{B}_K^{\text{WSR}}] &= \arg \min_{\mathbf{B}_1, \dots, \mathbf{B}_K} \sum_k -u_{R_k} R_k \\ \text{s.t. } &\sum_{k=1}^K \text{Tr}(\mathbf{B}_k \mathbf{B}_k^H) = E_{tx}, \end{aligned} \quad (4)$$

Where

- $u_{R_k} \geq 0$: weight for user k
- R_k : achievable rate for user k

$$R_k = \log \det (\mathbf{I}_k + \mathbf{B}_k^H \mathbf{H}_k^H \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k), \quad (5)$$

$$\mathbf{R}_{\tilde{v}_k \tilde{v}_k} = \mathbf{I}_k + \sum_{i=1, i \neq k}^K \mathbf{H}_k \mathbf{B}_i \mathbf{B}_i^H \mathbf{H}_k^H \quad (6)$$

The MMSE receive filter at user k is given by

$$\begin{aligned}\mathbf{A}_k^{\text{MMSE}} &= \arg \min_{\mathbf{A}_k} \mathbb{E} \left[\|\mathbf{A}_k \mathbf{y}_k - \mathbf{d}_k\|^2 \right] \\ &= \mathbf{B}_k^H \mathbf{H}_k^H \left(\mathbf{H}_k \mathbf{B}_k \mathbf{B}_k^H \mathbf{H}_k^H + \mathbf{R}_{\bar{v}_k \bar{v}_k} \right)^{-1}\end{aligned}\quad (7)$$

► Derivation

MSE Matrix for user k given at the MMSE-receive filter

$$\begin{aligned}\mathbf{E}_k &= \mathbb{E} \left[(\mathbf{A}_k^{\text{MMSE}} \mathbf{y}_k - \mathbf{d}_k) (\mathbf{A}_k^{\text{MMSE}} \mathbf{y}_k - \mathbf{d}_k)^H \right] \\ &= \left(\mathbf{I}_k + \mathbf{B}_k^H \mathbf{H}_k^H \mathbf{R}_{\bar{v}_k \bar{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k \right)^{-1}\end{aligned}\quad (8)$$

► Derivation

Given (5) and (8), the rate for user k can be expressed as

$$R_k = -\log \det(\mathbf{E}_k^{-1}) \quad (9)$$

Formulate Lagrangian for WSR maximization problem (4):

$$\begin{aligned} \{(\mathbf{B}_1, \dots, \mathbf{B}_K, \lambda) = & \underbrace{\sum_{k=1}^K -u_{R_k} R_k}_{\text{Objective: Minimize Negative WSR}} \\ & + \lambda \underbrace{\left(\sum_{k=1}^K \text{Tr}(\mathbf{B}_k \mathbf{B}_k^H) - E_{tx} \right)}_{\text{Constraint: Total Power Limit}} \end{aligned} \quad (12)$$

Define $\nabla_{\mathbf{B}_k} f = \frac{\partial f}{\partial \mathbf{B}_k^*}$

$$\begin{aligned}\nabla_{\mathbf{B}_k} f &= \nabla_{\mathbf{B}_k} \left(\sum_{j=1}^K -u_j R_j + \lambda \left(\sum_{j=1}^K \text{Tr}(\mathbf{B}_j \mathbf{B}_j^H) - E_{tx} \right) \right) \\ &= -u_k \underbrace{\nabla_{\mathbf{B}_k} R_k}_{\text{(i) Self-Rate}} - \sum_{i \neq k} u_i \underbrace{\nabla_{\mathbf{B}_k} R_i}_{\text{(ii) Interference}} + \lambda \mathbf{B}_k = \mathbf{0}\end{aligned}$$

Compute $\nabla_{\mathbf{B}_k} R_k$

Definition: $R_k = \log \det (\mathbf{I} + \mathbf{B}_k^H \mathbf{H}_k^H \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k)$

Gradient: $\nabla_{\mathbf{B}_k} R_k = \mathbf{H}_k^H \mathbf{R}_{\tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k \underbrace{(\mathbf{I} + \mathbf{B}_k^H \mathbf{H}_k^H \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k)^{-1}}_{\mathbf{E}_k \text{ (MMSE Matrix)}}$

$$\therefore \nabla_{\mathbf{B}_k} R_k = \mathbf{H}_k^H \mathbf{R}_{\tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k \mathbf{E}_k \quad (14)$$

► Derivation

Compute $\nabla_{\mathbf{B}_k} R_i$ for $i \neq k$

$$\nabla_{\mathbf{B}_k} R_i = -\mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_i \mathbf{E}_i \mathbf{B}_i^H \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_k \quad (17)$$

► Derivation

Combine (14) and (17), we obtain the gradient of WSR maximization problem (10):

$$\begin{aligned} \nabla_{\mathbf{B}_k} f &= -u_k \underbrace{\nabla_{\mathbf{B}_k} R_k}_{\text{(i) Self-Rate}} - \sum_{i \neq k} u_i \underbrace{\nabla_{\mathbf{B}_k} R_i}_{\text{(ii) Interference}} + \lambda \mathbf{B}_k \\ &= -u_k \mathbf{H}_k^H \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k \mathbf{E}_k + \left(\sum_{i=1, i \neq k}^K u_i \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_i \mathbf{E}_i \mathbf{B}_i^H \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \right) \mathbf{B}_k + \lambda \mathbf{B}_k \quad (10) \end{aligned}$$

The WMMSE minimization problem is given by

$$\begin{aligned} [\mathbf{B}_1^{\text{MMSE}}, \dots, \mathbf{B}_K^{\text{MMSE}}] &= \arg \min_{\mathbf{B}_1, \dots, \mathbf{B}_K} \sum_k \text{Tr}(\mathbf{W}_k \mathbf{E}_k) \\ \text{s.t. } \sum_{k=1}^K \text{Tr}(\mathbf{B}_k \mathbf{B}_k^H) &= E_{tx} \end{aligned} \quad (18)$$

$\mathbf{W}_k \in \mathbb{C}^{Q_k \times Q_k}$: constant weight matrix associated with user k .

Lagrangian leads:

$$g(\mathbf{B}_1 \cdots \mathbf{B}_K) = \sum_k \text{Tr}(\mathbf{W}_k \mathbf{E}_k) + \lambda \left(\sum_k \text{Tr}(\mathbf{B}_k \mathbf{B}_k^H) - E_{tx} \right) \quad (19)$$

Compute $\nabla_{\mathbf{B}_k} g$

1 Compute $\nabla_{\mathbf{B}_k} \text{Tr}(\mathbf{W}_k \mathbf{E}_k)$

$$\nabla_{\mathbf{B}_k} \text{Tr}(\mathbf{W}_k \mathbf{E}_k) = -\mathbf{H}_k^H \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k \mathbf{E}_k \mathbf{W}_k \mathbf{E}_k \quad (20)$$

► Derivation

2 Compute $\nabla_{\mathbf{B}_k} \text{Tr}(\mathbf{W}_i \mathbf{E}_i)$ for $i \neq k$

$$\nabla_{\mathbf{B}_k} \text{Tr}(\mathbf{W}_i \mathbf{E}_i) = \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_i \mathbf{E}_i \mathbf{W}_i \mathbf{E}_i \mathbf{B}_i^H \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_k \quad (21)$$

► Derivation

3 Combining (20) and (21), we obtain the WMMSE gradient:

$$\begin{aligned}
 \nabla_{\mathbf{B}_k} g &= \nabla_{\mathbf{B}_k} \text{Tr}(\mathbf{W}_k \mathbf{E}_k) + \left(\sum_{i=1, i \neq k}^K \nabla_{\mathbf{B}_k} \text{Tr}(\mathbf{W}_i \mathbf{E}_i) \right) + \lambda \mathbf{B}_k \\
 &= -\mathbf{H}_k^H \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k \mathbf{E}_k \mathbf{W}_k \mathbf{E}_k \\
 &\quad + \left(\sum_{i=1, i \neq k}^K \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_i \mathbf{E}_i \mathbf{W}_i \mathbf{E}_i \mathbf{B}_i^H \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \right) \mathbf{B}_k + \lambda \mathbf{B}_k \quad (11)
 \end{aligned}$$

Notice also

$$\nabla_{\lambda} f = \nabla_{\lambda} g = \sum_k \text{Tr}(\mathbf{B}_k \mathbf{B}_k^H) - E_{tx}$$

Key Insight: The two problems are closely related through the MSE weights.

- If we select the MSE weights as:

$$\mathbf{W}_k = u_k \mathbf{E}_k^{-1}$$

- Then, the gradients become identical:

$$\nabla_{\mathbf{B}_k} \text{WSR} = \nabla_{\mathbf{B}_k} \text{WMMSE}$$

Implication

A stationary point of the WSR problem satisfies the
KKT conditions of the WMMSE problem.

$\Rightarrow$ We can solve WSR by iteratively solving WMMSE.

Algorithm 1: WSRBF-WMMSE Algorithm

- 1 Set $n = 0$
 - 2 Set $\mathbf{B}_k^n = \mathbf{B}_k^{\text{init}} \forall k$
 - 3 **while** *not converged* **do**
 - 4 update $n \leftarrow n + 1$
 - 5 I. compute $\mathbf{A}_k^n \mid \mathbf{B}_i^{n-1} \forall i$ for all k using (7)
 - 6 II. compute $\mathbf{W}_k^n \mid \mathbf{B}_i^{n-1} \forall i$ for all k using (22),(8)
 - 7 III. compute $\mathbf{B}^n \mid \mathbf{A}^n, \mathbf{W}^n$ using (23),(24)
 - 8 **end**
-

The WMMSE transmit filter structure is computed as:

$$\bar{\mathbf{B}} = \left(\mathbf{H}^H \mathbf{A}^H \mathbf{W} \mathbf{A} \mathbf{H} + \frac{\text{Tr}(\mathbf{W} \mathbf{A} \mathbf{A}^H)}{E_{tx}} \mathbf{I}_P \right)^{-1} \mathbf{H}^H \mathbf{A}^H \mathbf{W}, \quad (23)$$

► Derivation

- $\mathbf{W}_{[QK \times QK]} = \text{diag}\{\mathbf{W}_1, \dots, \mathbf{W}_K\}$
- $\mathbf{A}_{[QK \times QK]} = \text{diag}\{\mathbf{A}_1, \dots, \mathbf{A}_K\}$
- $\mathbf{H}_{[QK \times P]} = [\mathbf{H}_1^T, \dots, \mathbf{H}_K^T]^T$: contains channel matrices stacked row-wise

$$\bar{\mathbf{B}} = \left(\underbrace{\mathbf{H}^H \mathbf{A}^H \mathbf{W} \mathbf{A} \mathbf{H}}_{\text{Signal Direction}} + \underbrace{\frac{\text{Tr}(\mathbf{W} \mathbf{A} \mathbf{A}^H)}{E_{tx}} \mathbf{I}_P}_{\text{Regularization Term}} \right)^{-1} \mathbf{H}^H \mathbf{A}^H \mathbf{W}$$

■ Regularization Term ($\mu \mathbf{I}_P$):

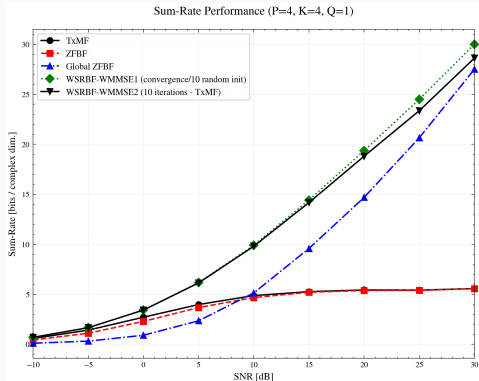
- The scalar factor $\mu = \frac{\text{Tr}(\mathbf{W} \mathbf{A} \mathbf{A}^H)}{E_{tx}}$ acts as a noise-to-signal ratio.
- Unlike standard ZFBF, this term prevents noise amplification when the channel is ill-conditioned.

■ Behavior at Extremes:

- **High SNR** ($E_{tx} \rightarrow \infty$): $\mu \rightarrow 0$, behaves like **Zero-Forcing (ZFBF)**.
- **Low SNR** ($E_{tx} \rightarrow 0$): μ dominates, behaves like **Matched Filter (TxMF)**.

Reproduce Setup

- $\mathbf{H}_k$: Rayleigh fading channel with path loss
- For all users, $u_{R_k} = 1$
- For matched filter (TxMF), $\mathbf{B}_k = \beta \mathbf{H}_k^H$
- For zero-forcing beamforming (ZFBF), $\mathbf{B} = \beta \mathbf{H}^H (\mathbf{H} \mathbf{H}^H)^{-1}$
- w/ Python, NumPy



Simulation Parameters: $P = 4, K = 4, Q = 1$

Performance Analysis:

- **Baseline Comparison:** TxMF and ZFBF are interference-limited (saturation) at high SNR.
- **WSRBF-WMMSE Success:** Shows linear growth w.r.t. SNR, indicating effective **interference management**.
- **Initialization Impact:** **WMMSE2** (TxMF Init + 10 iter) captures most of the gain of **WMMSE1** (Random Init + Convergence), verifying the low-complexity advantage.
- **Role of Weights (W):** The weight update acts as a **soft user selection**, nulling out users with poor channel compatibility to maximize sum-rate.

Code is available on GitHub

Thank You

Q & A

Derivation of Eq (3a)

$$\mathbf{x} = \sum_{k=1}^K \mathbf{B}_k \mathbf{d}_k, \quad \mathbf{x}^H \mathbf{x} = \left(\sum_{j=1}^K \mathbf{d}_j^H \mathbf{B}_j^H \right) \left(\sum_{k=1}^K \mathbf{B}_k \mathbf{d}_k \right) = \sum_{j=1}^K \sum_{k=1}^K \mathbf{d}_j^H \mathbf{B}_j^H \mathbf{B}_k \mathbf{d}_k$$

As $\mathbf{d}_k^H \mathbf{B}_k^H \mathbf{B}_k \mathbf{d}_k$ is scalar, Tr can be applied:

$$\mathbf{d}_j^H \mathbf{B}_j^H \mathbf{B}_k \mathbf{d}_k = \text{Tr}(\mathbf{B}_k^H \mathbf{B}_k \mathbf{d}_k \mathbf{d}_k^H), \quad \mathbb{E}[\mathbf{d}_j \mathbf{d}_k^H] = \begin{cases} \mathbf{I} & \text{if } j = k \\ 0 & \text{if } j \neq k \end{cases}$$

Therefore,

$$\mathbb{E}[\mathbf{x}^H \mathbf{x}] = \sum_{j=1}^K \sum_{k=1}^K \text{Tr}(\mathbf{B}_j^H \mathbf{B}_k \mathbb{E}[\mathbf{d}_k \mathbf{d}_j^H]) = \sum_{k=1}^K \text{Tr}(\mathbf{B}_k \mathbf{B}_k^H)$$

Derivation of Eq (7)

$$\begin{aligned} J &= \mathbb{E} \left[\|\mathbf{A}_k \mathbf{y}_k - \mathbf{d}_k\|^2 \right] = \mathbb{E} \left[\text{Tr} \left((\mathbf{A}_k \mathbf{y}_k - \mathbf{d}_k) (\mathbf{A}_k \mathbf{y}_k - \mathbf{d}_k)^H \right) \right] \\ &= \mathbb{E} \left[\text{Tr} \left((\mathbf{A}_k \mathbf{y}_k - \mathbf{d}_k) (\mathbf{y}_k^H \mathbf{A}_k^H - \mathbf{d}_k^H) \right) \right] \end{aligned}$$

Let Correlation $\mathbf{R}_{yy} = \mathbb{E} [\mathbf{y}_k \mathbf{y}_k^H]$, $\mathbf{R}_{dy} = \mathbb{E} [\mathbf{d}_k \mathbf{y}_k^H]$, $\mathbf{R}_{yd} = \mathbb{E} [\mathbf{y}_k \mathbf{d}_k^H] = \mathbf{R}_{dy}^H$

$$J = \text{Tr} (\mathbf{A}_k \mathbf{R}_{yy} \mathbf{A}_k^H - \mathbf{A}_k \mathbf{R}_{yd} - \mathbf{R}_{dy} \mathbf{A}_k^H + \mathbb{E} [\mathbf{d}_k \mathbf{d}_k^H])$$

Using matrix calculus,

$$\blacksquare \frac{\partial}{\partial \mathbf{X}^*} \text{Tr} (\mathbf{X} \mathbf{R} \mathbf{X}^H) = \mathbf{X} \mathbf{R}$$

$$\blacksquare \frac{\partial}{\partial \mathbf{X}^*} \cdot \text{Tr} (\mathbf{A} \mathbf{X}^H) = \mathbf{A}$$

$$\frac{\partial J}{\partial \mathbf{A}_k^*} = \mathbf{A}_k \mathbf{R}_{yy} - \mathbf{R}_{dy} = 0$$

$$\mathbf{A}_k^{\text{MMSE}} = \mathbf{R}_{dy} \mathbf{R}_{yy}^{-1}$$

Derivation of Eq (7)

$$\begin{aligned}\mathbf{R}_{dy} &= \mathbb{E} [\mathbf{d}_k \mathbf{y}_k^H] \\ &= \mathbb{E} [\mathbf{d}_k (\mathbf{H}_k \mathbf{B}_k \mathbf{d}_k + \tilde{\mathbf{v}}_k)^H] \\ &= \mathbb{E} [\mathbf{d}_k \mathbf{d}_k^H \mathbf{B}_k^H \mathbf{H}_k^H] = \mathbb{E} [\mathbf{d}_k \mathbf{d}_k^H] \mathbf{B}_k^H \mathbf{H}_k^H \\ &= \mathbf{B}_k^H \mathbf{H}_k^H\end{aligned}$$

$$\begin{aligned}\mathbf{R}_{yy} &= \mathbb{E} [\mathbf{y}_k \mathbf{y}_k^H] \\ &= \mathbb{E} [(\mathbf{H}_k \mathbf{B}_k \mathbf{d}_k + \tilde{\mathbf{v}}_k) (\mathbf{H}_k \mathbf{B}_k \mathbf{d}_k + \tilde{\mathbf{v}}_k)^H] \\ &= \mathbf{B}_k \mathbf{H}_k \mathbb{E} [\mathbf{d}_k \mathbf{d}_k^H] \mathbf{B}_k^H \mathbf{H}_k^H + \mathbb{E} [\tilde{\mathbf{v}}_k \tilde{\mathbf{v}}_k^H] \\ &= \mathbf{B}_k \mathbf{H}_k \mathbf{B}_k^H \mathbf{H}_k^H + \mathbf{R}_{\tilde{v}_k \tilde{v}_k}\end{aligned}$$

Therefore,

$$\mathbf{A}_k^{\text{MMSE}} = \underbrace{\mathbf{B}_k^H \mathbf{H}_k^H}_{\mathbf{R}_{dy}} \underbrace{(\mathbf{H}_k \mathbf{B}_k \mathbf{B}_k^H \mathbf{H}_k^H + \mathbf{R}_{\tilde{v}_k \tilde{v}_k})^{-1}}_{\mathbf{R}_{yy}^{-1}}$$

Derivation of Eq (8)

$$\begin{aligned}\mathbf{E}_k &= \mathbb{E} \left[\left(\mathbf{A}_k^{\text{MMSE}} \mathbf{y}_k - \mathbf{d}_k \right) \left(\mathbf{A}_k^{\text{MMSE}} \mathbf{y}_k - \mathbf{d}_k \right)^H \right] \\&= \mathbf{A}_k \mathbf{R}_{yy} \mathbf{A}_k^H - \mathbf{A}_k \mathbf{R}_{yd} - \mathbf{R}_{dy} \mathbf{A}_k^H + \mathbf{R}_{dd} \\&= (\mathbf{R}_{dy} \mathbf{R}_{yy}^{-1}) \mathbf{R}_{yy} \mathbf{A}_k^H - (\mathbf{R}_{dy} \mathbf{R}_{yy}^{-1}) \mathbf{R}_{yd} - \mathbf{R}_{dy} \mathbf{A}_k^H + \mathbf{R}_{dd} \\&= \mathbf{R}_{dd} - \mathbf{R}_{dy} \mathbf{R}_{yy}^{-1} \mathbf{R}_{yd} \\&= \mathbf{I} - \mathbf{B}_k^H \mathbf{H}_k^H \left(\mathbf{B}_k \mathbf{H}_k \mathbf{B}_k^H \mathbf{H}_k^H + \mathbf{R}_{\tilde{v}_k \tilde{v}_k} \right)^{-1} \mathbf{H}_k \mathbf{B}_k\end{aligned}$$

Apply Woodbury matrix identity $\mathbf{I} - \mathbf{U}(\mathbf{C} + \mathbf{V}\mathbf{U})^{-1}\mathbf{V} = (\mathbf{I} + \mathbf{U}\mathbf{C}^{-1}\mathbf{V})^{-1}$,
Let $\mathbf{U} = \mathbf{B}_k^H \mathbf{H}_k^H$, $\mathbf{V} = \mathbf{H}_k \mathbf{B}_k$, $\mathbf{C} = \mathbf{R}_{\tilde{v}_k \tilde{v}_k}$

$$\begin{aligned}\mathbf{E}_k &= \left(\mathbf{I} + \left(\mathbf{B}_k^H \mathbf{H}_k^H \right) \left(\mathbf{R}_{\tilde{v}_k \tilde{v}_k} \right)^{-1} \left(\mathbf{H}_k \mathbf{B}_k \right) \right)^{-1} \\&= \left(\mathbf{I} + \mathbf{B}_k^H \mathbf{H}_k^H \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k \right)^{-1}\end{aligned}$$

Derivation of Eq (14)

Target: $R_k = \log \det(\mathbf{K})$

Substitution: $\mathbf{K} = \mathbf{I} + \mathbf{B}_k^H \mathbf{C} \mathbf{B}_k$

Constant Matrix: $\mathbf{C} = \mathbf{H}_k^H \mathbf{R}_{\tilde{v}_k}^{-1} \mathbf{H}_k$

MMSE Matrix: $\mathbf{E}_k = \mathbf{K}^{-1}$

$$\frac{\partial \log \det \mathbf{K}}{\partial x} = \text{Tr}(\mathbf{K}^{-1} \frac{\partial \mathbf{K}}{\partial x})$$

$$\frac{\partial R_k}{\partial [\mathbf{B}_k^*]_{nm}} = \text{Tr} \left(\mathbf{K}^{-1} \frac{\partial \mathbf{K}}{\partial [\mathbf{B}_k^*]_{nm}} \right) = \text{Tr} \left(\mathbf{E}_k \frac{\partial \mathbf{K}}{\partial [\mathbf{B}_k^*]_{nm}} \right)$$

$$\frac{\partial \mathbf{K}}{\partial [\mathbf{B}_k^*]_{nm}} = \mathbf{e}_m \mathbf{e}_n^T \mathbf{C} \mathbf{B}_k$$

$$\begin{aligned} \frac{\partial R_k}{\partial [\mathbf{B}_k^*]_{nm}} &= \text{Tr} (\mathbf{E}_k \mathbf{e}_m \mathbf{e}_n^T \mathbf{C} \mathbf{B}_k) \\ &= \text{Tr} (\mathbf{e}_n^T \mathbf{C} \mathbf{B}_k \mathbf{E}_k \mathbf{e}_m) \\ &= [\mathbf{C} \mathbf{B}_k \mathbf{E}_k]_{nm} \end{aligned}$$

$$\nabla_{\mathbf{B}_k} R_k = \mathbf{C} \mathbf{B}_k \mathbf{E}_k = \mathbf{H}_k^H \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k \mathbf{E}_k$$

Derivation of Eq (17)

$$\mathbf{S} = \mathbf{X} + \mathbf{L}\mathbf{K}\mathbf{C}\mathbf{C}^{\mathbf{H}}\mathbf{K}^{\mathbf{H}}\mathbf{L}^{\mathbf{H}}$$

$$\nabla_{\mathbf{K}}h = \mathbf{L}^{\mathbf{H}} (\nabla_{\mathbf{S}}h) \mathbf{L}\mathbf{K}\mathbf{C}\mathbf{C}^{\mathbf{H}}$$

$$R_i = \log \det (\mathbf{I} + \mathbf{B}_i^{\mathbf{H}}\mathbf{H}_i^{\mathbf{H}}\mathbf{R}_{\tilde{v}_i\tilde{v}_i}^{-1}\mathbf{H}_i\mathbf{B}_i)$$

$$\mathbf{R}_{\tilde{v}_i\tilde{v}_i} = \mathbf{I}_i + \sum_{k=1, k \neq i}^K \mathbf{H}_i\mathbf{B}_k\mathbf{B}_k^{\mathbf{H}}\mathbf{H}_i^{\mathbf{H}}$$

$$\mathbf{R}_{\tilde{v}_i\tilde{v}_i} = \underbrace{\left(\mathbf{I}_i + \sum_{\substack{j=1 \\ j \neq i, k}}^K \mathbf{H}_i\mathbf{B}_j\mathbf{B}_j^{\mathbf{H}}\mathbf{H}_i^{\mathbf{H}} \right)}_{\mathbf{X} \text{ (constant)}} + \mathbf{H}_i\mathbf{B}_k\mathbf{I}\mathbf{B}_k^{\mathbf{H}}\mathbf{H}_i^{\mathbf{H}}$$

$$h = R_i, \quad \mathbf{S} = \mathbf{R}_{\tilde{v}_i\tilde{v}_i}, \quad \mathbf{L} = \mathbf{H}_i, \quad \mathbf{K} = \mathbf{B}_k, \quad \mathbf{C} = \mathbf{C}^{\mathbf{H}} = \mathbf{I}$$

$$\nabla_{\mathbf{B}_k}R_i = \mathbf{H}_i^{\mathbf{H}} (\nabla_{\mathbf{R}_{\tilde{v}_i\tilde{v}_i}}R_i) \mathbf{H}_i\mathbf{B}_k \quad (15)$$

Derivation of Eq (17)

1. Definitions

$$R_i = \log \det(\mathbf{K}_i)$$

$$\mathbf{K}_i = \mathbf{I} + \mathbf{A}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{A}_i \quad (\text{where } \mathbf{A}_i = \mathbf{H}_i \mathbf{B}_i)$$

$$\mathbf{E}_i = \mathbf{K}_i^{-1} \quad (\text{MMSE Matrix})$$

2. Differential of Log-Det (Chain Rule)

$$dR_i = \text{Tr}(\mathbf{K}_i^{-1} d\mathbf{K}_i) = \text{Tr}(\mathbf{E}_i d\mathbf{K}_i)$$

3. Differential of Inverse Matrix

Using $d(\mathbf{X}^{-1}) = -\mathbf{X}^{-1}(d\mathbf{X})\mathbf{X}^{-1}$:

$$\begin{aligned} d\mathbf{K}_i &= \mathbf{A}_i^H d(\mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1}) \mathbf{A}_i \\ &= \mathbf{A}_i^H \left(-\mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} (d\mathbf{R}_{\tilde{v}_i \tilde{v}_i}) \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \right) \mathbf{A}_i \end{aligned}$$

4. Substitution & Trace Cyclic Property

$$dR_i = \text{Tr} \left(\mathbf{E}_i \left[-\mathbf{A}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} (d\mathbf{R}_{\tilde{v}_i \tilde{v}_i}) \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{A}_i \right] \right)$$

Move $(d\mathbf{R}_{\tilde{v}_i \tilde{v}_i})$ to the end:

$$dR_i = \text{Tr} \left(\underbrace{\left[-\mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{A}_i \mathbf{E}_i \mathbf{A}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \right]}_{\text{Gradient}} d\mathbf{R}_{\tilde{v}_i \tilde{v}_i} \right)$$

5. Obtain Eq (16)

$$\nabla_{\mathbf{R}_{\tilde{v}_i \tilde{v}_i}} R_i = -\mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_i \mathbf{E}_i \mathbf{B}_i^H \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1}$$

Derivation of Eq (17)

Combining Eq (15) and (16), we obtain Eq (17):

$$\nabla_{\mathbf{R}_{\tilde{v}_i \tilde{v}_i}} R_i = -\mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_i \mathbf{E}_i \mathbf{B}_i^H \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \quad (16)$$

$$\nabla_{\mathbf{B}_k} R_i = \mathbf{H}_i^H (\nabla_{\mathbf{R}_{\tilde{v}_i \tilde{v}_i}} R_i) \mathbf{H}_i \mathbf{B}_k \quad (15)$$

$$= -\mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_i \mathbf{E}_i \mathbf{B}_i^H \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_k \quad (17)$$

Derivation of Eq (20)

$$\mathbf{E}_k = \left(\mathbf{I}_k + \mathbf{B}_k^H \underbrace{\mathbf{H}_k^H \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k}_{\mathbf{F}} \mathbf{B}_k \right)^{-1}$$

$$d(\mathbf{X}^{-1}) = -\mathbf{X}^{-1} (d\mathbf{X}) \mathbf{X}^{-1},$$

$$\begin{aligned} d(\mathbf{E}_k) &= -\mathbf{E}_k (d\mathbf{E}_k^{-1}) \mathbf{E}_k \\ &= -\mathbf{E}_k (d(\mathbf{I}_k + \mathbf{B}_k^H \mathbf{F} \mathbf{B}_k)) \mathbf{E}_k, \\ d(\mathbf{B}_k^H \mathbf{F} \mathbf{B}_k) &= (d\mathbf{B}_k^H) \mathbf{F} \mathbf{B}_k \end{aligned}$$

$$\begin{aligned} d \operatorname{Tr}(\mathbf{W}_k \mathbf{E}_k) &= \operatorname{Tr}(\mathbf{W}_k (d\mathbf{E}_k)) \\ &= \operatorname{Tr}(\mathbf{W}_k (-\mathbf{E}_k (d\mathbf{B}_k^H) \mathbf{F} \mathbf{B}_k \mathbf{E}_k)) \\ &= -\operatorname{Tr}(\mathbf{W}_k \mathbf{E}_k (d\mathbf{B}_k^H) \mathbf{F} \mathbf{B}_k \mathbf{E}_k) \\ &= -\operatorname{Tr} \left(\underbrace{\mathbf{F} \mathbf{B}_k \mathbf{E}_k \mathbf{W}_k \mathbf{E}_k}_{\nabla_{\mathbf{B}_k} \operatorname{Tr}(\mathbf{W}_k \mathbf{E}_k)} (d\mathbf{B}_k^H) \right) \end{aligned}$$

$$\therefore \nabla_{\mathbf{B}_k} \operatorname{Tr}(\mathbf{W}_k \mathbf{E}_k) = -\mathbf{H}_k^H \mathbf{R}_{\tilde{v}_k \tilde{v}_k}^{-1} \mathbf{H}_k \mathbf{B}_k \mathbf{E}_k \mathbf{W}_k \mathbf{E}_k \quad (20)$$

Derivation of Eq (21)

$$\mathbf{E}_i = (\mathbf{I}_i + \mathbf{B}_i^H \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_i)^{-1}$$

$\mathbf{B}_k$ is only relevant with $\mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1}$

$$\begin{aligned} d(\mathbf{E}_i) &= -\mathbf{E}_i (d\mathbf{E}_i^{-1}) \mathbf{E}_i \\ &= -\mathbf{E}_i (d(\mathbf{I}_i + \mathbf{B}_i^H \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_i)) \mathbf{E}_i \\ &= -\mathbf{E}_i (\mathbf{B}_i^H \mathbf{H}_i^H (d\mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1}) \mathbf{H}_i \mathbf{B}_i) \mathbf{E}_i \end{aligned}$$

$$d(\mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1}) = -\mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} (d\mathbf{R}_{\tilde{v}_i \tilde{v}_i}) \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1}$$

$$\begin{aligned} d \text{Tr}(\mathbf{W}_i \mathbf{E}_i) &= \text{Tr}(\mathbf{W}_i (d\mathbf{E}_i)) \\ &= -\text{Tr}(\mathbf{W}_i \mathbf{E}_i \mathbf{B}_i^H \mathbf{H}_i^H (d\mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1}) \mathbf{H}_i \mathbf{B}_i \mathbf{E}_i) \\ &= -\text{Tr}(\mathbf{H}_i \mathbf{B}_i \mathbf{E}_i \mathbf{W}_i \mathbf{E}_i \mathbf{B}_i^H \mathbf{H}_i^H (d\mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1})) \\ &= \text{Tr}(\mathbf{H}_i \mathbf{B}_i \mathbf{E}_i \mathbf{W}_i \mathbf{E}_i \mathbf{B}_i^H \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} (d\mathbf{R}_{\tilde{v}_i \tilde{v}_i}) \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1}) \\ &= \text{Tr} \left(\underbrace{\mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_i \mathbf{E}_i \mathbf{W}_i \mathbf{E}_i \mathbf{B}_i^H \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1}}_{\nabla_{\mathbf{R}_{\tilde{v}_i \tilde{v}_i}} \text{Tr}(\mathbf{W}_i \mathbf{E}_i)} (d\mathbf{R}_{\tilde{v}_i \tilde{v}_i}) \right) \end{aligned}$$

Derivation of Eq (21)

$$\mathbf{R}_{\tilde{v}_i \tilde{v}_i} = \cdots + \mathbf{H}_i \mathbf{B}_k \mathbf{B}_k^H \mathbf{H}_i^H + \cdots$$

$$d\mathbf{R}_{\tilde{v}_i \tilde{v}_i} = \mathbf{H}_i \mathbf{B}_k (d\mathbf{B}_k^H) \mathbf{H}_i^H$$

$$\begin{aligned} d \operatorname{Tr}(\mathbf{W}_i \mathbf{E}_i) &= \operatorname{Tr}(\alpha (d\mathbf{R}_{\tilde{v}_i \tilde{v}_i})) \\ &= \operatorname{Tr}(\alpha \mathbf{H}_i \mathbf{B}_k (d\mathbf{B}_k^H) \mathbf{H}_i^H) \\ &= \operatorname{Tr} \left(\underbrace{\mathbf{H}_i^H \alpha \mathbf{H}_i \mathbf{B}_k}_{\nabla_{\mathbf{B}_k} \operatorname{Tr}(\mathbf{W}_i \mathbf{E}_i)} (d\mathbf{B}_k^H) \right) \end{aligned}$$

$$\nabla_{\mathbf{B}_k} \operatorname{Tr}(\mathbf{W}_i \mathbf{E}_i) = \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_i \mathbf{E}_i \mathbf{W}_i \mathbf{E}_i \mathbf{B}_i^H \mathbf{H}_i^H \mathbf{R}_{\tilde{v}_i \tilde{v}_i}^{-1} \mathbf{H}_i \mathbf{B}_k \quad (21)$$

Derivation of Eq (23)

Problem Formulation: Minimize the Weighted MSE subject to the transmit power constraint E_{tx} . Let $\mathbf{B} = b\bar{\mathbf{B}}$, where b is a scaling factor.

$$\begin{aligned} \min_{\bar{\mathbf{B}}, b} \quad & J = \mathbb{E} \left[\left\| \mathbf{W}^{1/2}(\mathbf{s} - b^{-1}\mathbf{A}\mathbf{y}) \right\|^2 \right] \\ \text{s.t.} \quad & \text{Tr}(\mathbf{B}\mathbf{B}^H) = b^2 \text{Tr}(\bar{\mathbf{B}}\bar{\mathbf{B}}^H) = E_{tx} \end{aligned}$$

Step 1: Constraint Substitution

From the constraint, we substitute the scaling factor b :

$$b^{-2} = \frac{\text{Tr}(\bar{\mathbf{B}}\bar{\mathbf{B}}^H)}{E_{tx}}$$

Step 2: Cost Function Expansion

Substituting $\mathbf{y} = \mathbf{H}b\bar{\mathbf{B}}\mathbf{s} + \mathbf{n}$ and b^{-2} into the cost function J :

$$\begin{aligned} J(\bar{\mathbf{B}}) &= \text{Tr} \left(\mathbf{W}(\mathbf{I} - \mathbf{A}\mathbf{H}\bar{\mathbf{B}})(\mathbf{I} - \mathbf{A}\mathbf{H}\bar{\mathbf{B}})^H \right) + b^{-2} \text{Tr}(\mathbf{W}\mathbf{A}\mathbf{A}^H) \\ &= \underbrace{\text{Tr} \left(\mathbf{W}(\mathbf{I} - \mathbf{A}\mathbf{H}\bar{\mathbf{B}})(\dots)^H \right)}_{\text{ErrorTerm}} + \underbrace{\frac{\text{Tr}(\mathbf{W}\mathbf{A}\mathbf{A}^H)}{E_{tx}} \text{Tr}(\bar{\mathbf{B}}\bar{\mathbf{B}}^H)}_{\text{PenaltyTerm(fromNoise)}} \end{aligned}$$

Derivation of Eq (23)

Step 3: Complex Gradient

Taking the gradient with respect to $\bar{\mathbf{B}}^*$ and setting it to zero:

$$\frac{\partial J}{\partial \bar{\mathbf{B}}^*} = -\mathbf{H}^H \mathbf{A}^H \mathbf{W} + \mathbf{H}^H \mathbf{A}^H \mathbf{W} \mathbf{A} \mathbf{H} \bar{\mathbf{B}} + \underbrace{\frac{\text{Tr}(\mathbf{W} \mathbf{A} \mathbf{A}^H)}{E_{tx}} \mathbf{I}_P}_{\text{Regularization}} \bar{\mathbf{B}} = 0$$

Step 4: Solution

Rearranging the terms to solve for $\bar{\mathbf{B}}$:

$$\left(\mathbf{H}^H \mathbf{A}^H \mathbf{W} \mathbf{A} \mathbf{H} + \frac{\text{Tr}(\mathbf{W} \mathbf{A} \mathbf{A}^H)}{E_{tx}} \mathbf{I}_P \right) \bar{\mathbf{B}} = \mathbf{H}^H \mathbf{A}^H \mathbf{W}$$

Finally, we obtain the structure in (23):

$$\bar{\mathbf{B}} = \left(\mathbf{H}^H \mathbf{A}^H \mathbf{W} \mathbf{A} \mathbf{H} + \frac{\text{Tr}(\mathbf{W} \mathbf{A} \mathbf{A}^H)}{E_{tx}} \mathbf{I}_P \right)^{-1} \mathbf{H}^H \mathbf{A}^H \mathbf{W}$$